Aroudra Thakur

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Summary

CS Honors student (expected May 2027) with three industry internships building **LLM agents, RAG systems, AI observability, and full-stack products**, undergraduate research in event-based vision, pose estimation, and ASL recognition, and experience leading student engineering teams of 5 and 7.

Education

University of Texas at Arlington *Expected May 2027*

B.S. in Computer Science (Honors) | **GPA: 3.86/4.00** | Minor: Mathematics

Fundamentals of AI Certificate Program | Leadership Honors Program | Maverick Academic Scholarship ($100,000 over four years)

Relevant Coursework: Artificial Intelligence, Machine Learning, Computer Vision, Neural Networks & Deep Learning, Algorithms, Data Structures, Linear Algebra, Probability, Robotics, Operating Systems

Experience

Artificial Intelligence Intern *Jun 2026 – Aug 2026*

HF Sinclair *Dallas, TX*

- Developed **Python/LangGraph multi-agent orchestration** for an enterprise contract-analysis workflow, owning document ingestion, semantic retrieval (**Snowflake Cortex Search**), tool design, routing, shared structured state, guardrails, validation, and failure handling across specialized agents.
- Created **ObsFlow**, a reusable Snowflake-backed observability and evaluation framework capturing agent traces, spans, tool calls, latency, citation provenance, quality signals, and failures for root-cause analysis.
- Contributed to an internal **HR AI assistant** using Snowflake Cortex Agents, Cortex Search, SQL, and dbt-backed organizational data; developed question routing, grounding, citation handling, and retrieval evaluation.
- Built an **80-question evaluation suite** and reduced measured token consumption by approximately **70%** (20K to 6K tokens per request).
- Traced a citation-generation failure to an incompatible Snowflake function implementation, corrected the execution path, and presented results to engineers and business stakeholders in team reviews and demonstrations.

AI Engineering Intern *Dec 2025 – May 2026*

Ozcorp Scientific LLC *Remote (Guaynabo, PR)*

- Developed a full-stack **AI research agent** (Python, FastAPI, React, embeddings, semantic search) automating company resolution, crawl/search discovery, structured extraction, and report generation; reduced manual research work by **80%+**.
- Designed hybrid retrieval, entity disambiguation, and ranking workflows; improved result quality by **30%** and reduced irrelevant results by **35%**.
- Implemented link verification and LLM-based relevance scoring for executive, lawsuit, and press-release attribution, reducing false positives by **60%+** and improving extraction precision by **40%**.
- Added source verification, validation, and fallback handling; cut end-to-end latency by **50%**.

Software Engineering Intern *Aug 2024 – Dec 2024*

Hotspring *Remote (Barrington, NH)*

- Developed and deployed production **React/TypeScript** and **Django/GraphQL** features on a live web platform using PostgreSQL and AWS S3.
- Built a Media Shuttle API and email-parsing service (Celery workers) that returns S3 file locations by automated reply, cutting manual file search by **95%+**.
- Added a Project Category Type module with GraphQL refetching (UI responsiveness up **80%**) and redesigned invoice management with view/download features.

Research Experience

Undergraduate Research Assistant, Event-Based Vision *May 2025 – Present*

PRIMAL Lab, University of Texas at Arlington *Arlington, TX*

- Built Python/PyTorch pipelines for asynchronous event-camera streams; compared four event representations and trained **ResNet-18** for steering-angle prediction on DDD20 (**32.8° RMSE**).
- Collaborating with Ph.D. students on a heat-diffusion event representation for machine learning models under Prof. Diego Patiño; authored the research poster "Heat-Diffusion Modeling of Event Streams."
- Evaluated event representations on EVIMO, explored loss functions on DDD17 with CNNs and ResNet-50, and created an open-source GitHub tool converting raw (x, y, t, p) event data into visual representations across datasets.

Undergraduate Researcher, Human Pose Estimation *Oct 2024 – Oct 2025*

University of Texas at Arlington *Arlington, TX*

- Coauthored a peer-reviewed survey of **2D/3D human pose estimation** (CNNs, Transformers, graph-based methods, benchmarks, metrics, deployment constraints) and contributed to dataset preprocessing and pipeline comparisons in PyTorch, TensorFlow, and OpenCV.

Undergraduate Researcher, ASL Hand-Pose Similarity *Aug 2024 – Aug 2025*

University of Texas at Arlington *Arlington, TX*

- Implemented **dynamic time warping, edit distance, Chamfer distance, and shape-context matching** to compare hand

poses and gesture trajectories in a nearest-neighbor benchmark on the NYU Hand Pose dataset; built OpenCV/MediaPipe keypoint normalization pipelines and correspondence visualization tools.

Publications

Journal article (equal-contributing coauthor). R. Shrestha, **A. S. Thakur**, and C. Wang, "Human Pose Estimation in 2D and 3D: A Survey of Analytical Methods, Benchmarking Frameworks, and Engineering Applications," *Journal of Experimental and Theoretical Analyses*, vol. 4, no. 3, art. 28, 2026. DOI: 10.3390/jeta4030028.

Conference paper (sole author, presented as a poster). A. S. Thakur, "Assistive Technologies for Learning Disabilities: A Systematic Review of Trends and Impact," 2025 ASEE Gulf-Southwest Annual Conference, Arlington, TX, Mar. 2025. DOI: 10.18260/1-2–55030.

Leadership & Teaching

Director of Technology *Jul 2026 – Nov 2026*
HackUTA *Arlington, TX*

- Led **5 engineers** building HackUTA's application and event platform in React, TypeScript, Vite, and Convex, built to support 1,000+ applications; owned task assignment, architecture decisions, pull-request review, and deployments.
- Implemented password and email-OTP authentication, **role-based access control**, server-side authorization limiting applicants to their own data, and audit logs for application-review decisions.
- Hardened the platform with CSP, authentication and upload rate limits, shared Zod validation, input sanitization, and secure resume uploads (origin allowlists, PDF validation, single-use tokens).
- Built asynchronous email queues with delivery-status tracking and service health endpoints, and set up **GitHub Actions CI/CD** with Vitest, Playwright end-to-end tests, and 90% coverage gates.

Project Manager, Silo *Jan 2026 – Present*
ACM (Create Committee), University of Texas at Arlington *Arlington, TX*

- Lead **7 engineers** building an LLM prompt-regression testing platform; run sprint planning and milestones and architected the core drift-testing pipeline.

Supplemental Instruction Leader *Jan 2025 – Dec 2025*
University of Texas at Arlington *Arlington, TX*

- Led peer study sessions in Data Structures & Algorithms and Discrete Structures; created problem sets, mock exams, and review materials.

Selected Projects

Silo: LLM Evaluation Platform | *Python, FastAPI, Next.js, Supabase | GitHub | Live | npm CLI*

- Built prompt drift detection comparing baseline and candidate versions through staged evaluations over stored test cases, with semantic similarity scoring (embeddings, 0–1 scale), latency tracking, and CI-ready pass/fail gating; evaluation accuracy improved by approximately 35%.
- Built FastAPI/Supabase services storing versioned prompt suites and 1K+ test executions, a Next.js dashboard with drift summaries, and a published npm CLI with GitHub Actions integration.

Cicada AML: Graph-Based Risk Analysis | *Python, PyTorch Geometric, XGBoost, FastAPI, React | GitHub*

- Built an ML risk-analysis app for blockchain transactions that builds wallet graphs and combines **185 heuristics** with **five ML models** (GNN, LSTM, XGBoost, and others) trained on the Elliptic Bitcoin dataset (203K+ transactions).
- Added an XGBoost meta-learner to calibrate heuristic and model signals into a probabilistic risk score, graph attention and Louvain/Leiden community detection for laundering networks, and SHAP explanations.
- Built a React dashboard (Cytoscape.js, Plotly) for transaction flows, suspicious clusters, reports, and SAR exports.

AutoResume: AI Resume Generation Platform | *Next.js, FastAPI, pgvector, Celery, AWS Cognito | GitHub*

- Built pgvector matching of a structured career profile to job descriptions, GPT-based optimization, and deterministic LaTeX rendering (Tectonic), cutting manual resume editing time by approximately 80%.
- Added Celery/Redis asynchronous PDF/DOCX generation, AWS Cognito authentication, and AES-256-GCM encryption of sensitive user data, with 80%+ backend and 70%+ frontend test coverage.

Nixo: Real-Time Support Triage | *TypeScript, Next.js, OpenAI, Supabase, Socket.IO | GitHub*

- Built a Slackbot and live Socket.IO dashboard that classifies messages with OpenAI and groups them into tickets using thread context, entity extraction, vector similarity, and LLM validation (approximately 60% fewer duplicate tickets), with sequential processing to keep ticket state consistent under concurrency.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C/C++, PHP, Kotlin
AI/ML: LangGraph, LangChain, RAG, Embeddings, Vector Search (pgvector), Snowflake Cortex, Agent Evaluation & Observability, Prompt Engineering, PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, XGBoost, SHAP, OpenCV, MediaPipe
Backend & Data: FastAPI, Django, Node.js, REST, GraphQL, PostgreSQL, Supabase, MySQL, Snowflake, dbt, Redis, Celery, Convex, Socket.IO; auth/security (Email OTP, RBAC, rate limiting, CSP, audit logging)
Frontend & Infrastructure: React, Next.js, Angular, Tailwind CSS, AWS (S3, Cognito), Docker, Git, GitHub Actions, CI/CD, Pytest, Vitest, Playwright

Honors & Awards

Alpha Lambda Delta | Honors College | President's List (Spring 2026) | Dean's List (Fall 2025) | Freshman Distinction Roll (Fall - Spring 2023) | UTA Table Tennis Captain